Caleb Chelstrom

Anoka, MN | cachelstrom@gmail.com

DevSecOps Engineer

Solutions Engineering | Automation | IT Architecture

Senior DevSecOps Engineer with extensive experience in automating policy-as-code and implementing innovative solutions in high-performing DevSecOps/Middleware environments. Proficient in python object-oriented programming, security operations engineering, API development, IAM enterprise solution onboarding and integrations, technical design, and enterprise-level automation software development. Seeking opportunities to contribute expertise in engineering to drive efficiency and innovation in a dynamic organization.

Key Skills

Python | MySQL | Red Hat Enterprise Linux | Git | Git Bash | Jenkins | Kubernetes | Rancher | Terraform | Ansible | Azure | AWS | HashiCorp Vault | SailPoint (IIQ) | REST APIs | API Development & Testing | Postman | ServiceNow | PowerShell | Active Directory | RBAC | EBAC | PBAC | Apache Airflow | GitLab | Jira | Solution Architecture | Technical Solution Design | Process Flow Diagram (PFD) | Agile | Object-Oriented Programming | Identity & Access Management (IAM)

Work Experience

Senior Integration Engineer

U.S. Bank | St. Paul, MN 2024 - Present

- Sole SME responsible for maintaining the configuration and integration of the Bank's HashiCorp Vault Enterprise solution for all application use cases.
- Led the seamless onboarding and integration of Identity and Access Management (IAM) software solutions into customer software ecosystems, enhancing system interoperability and user access control through OAuth and SAML protocols, ensuring GDPR compliance.
- Pioneer integrations of HashiCorp Vault for all U.S. bank application resulting in 28,000+ stored secrets across all namespaces, and roughly 150,000 authentication tokens generated per hour.
- Manage custom integration use cases of various OAuth solutions to HashiCorp Vault for applications to generate tokens or retrieve secrets via retrievable JWT (JSON Web Token), Java client connections for Java applications, Kubernetes service accounts, various methods of AWS user authentication (CLI, UI, Python, cURL), and authentication via a GitLab Project/ Group and a GitLab Runner, and finally a home built auto auth solution for Windows and Linux VMs.
- Maintain and enhance automated and manual process pertaining to HashiCorp Vault entity alias policy assignment and customization, engine provisioning, and secret rotation.
- Utilize Terraform for managing IAM permissions on our AWS account and for provisioning our S3 buckets for backing up the past one-hundred versions of HashiCorp Vault.
- Own and manage API onboarding and integrations for all IAM automation customers, ensuring least privilege via JSON based API policies for specific application solution use cases for our APIs that perform read write actions with SailPoint IIQ, Active Directory, CyberArk PAM, HashiCorp Vault, SQL databases, and RHEL virtual machines.
- Consult internal customers on U.S. Bank best practices outside of our immediate security operational scope such as certificate authority, and key management.

Software Engineer

U.S. Bank | St. Paul, MN 2022 – 2024

- Pioneered the automation of RBAC, EBAC, and PBAC via policy-as-code and identity access frameworks across over 60 JIRA stories, leveraging InfoSec and non-security application teams, significantly enhancing operational efficiencies through streamlined GDPR-compliant processes.
- Conceptualized and implemented a cutting-edge REST Microservice Solution, engineered 13 APIs from scratch with Swagger for documentation, this suite of APIs perform various read write operations against LDAP, MySQL database, and CyberArk that revolutionized operations by obviating manual server access provisioning previously handled by the entire Unix/Linux System Administration team, thereby enhancing productivity and system reliability all new RHEL virtual machine builds.
- Coordinated cross-functional teams in the development and deployment of Identity Management solutions into their own IaC solutions (Ansible, Terraform, Helm).
- Fostered collaborative efforts with Security Access Operations teams to devise a user-centric, self-service platform for Active Directory object management, employing Python scripting to achieve a groundbreaking 100% reduction in manual task loads, substantially improving operational agility and SOC 2 compliance.
- Orchestrated an innovative object-oriented strategy, using Python scripts, to systematically revoke access for users over-due on quarterly security training, achieving remarkable cost savings of $5.75 million by enhancing overall Identity Lifecycle Management and enhancing compliance with industry standards like ISO 27001.
- Innovated and deployed a sophisticated GUI leveraging REST APIs for the remote execution of CLI administrative tasks on RedHat Linux servers, utilizing React for the front end, securing $8.3 thousand in cost efficiencies and doubling process efficiency, thereby setting a new standard for server provisioning workflows.
- Refactored OIM related automation code and REST APIs in preparation for migrating our access management solution to SailPoint.

Unix/Linux System Administrator
U.S. Bank | St. Paul, MN 2021 – 2022

- Elevated Info Security's SLA attainment to 99% for 2021/2022, by leading the strategic overhaul of access management processes, accommodating over 60,000 onsite and remote users with GDPR and SOC 2 compliant protocols, thereby ensuring robust security and operational excellence.
- As the Unix/Linux Subject Matter Expert (SME), fostered a synergistic collaboration with application and database teams using Agile methodologies, markedly advancing system access management efficiency across all server platforms.
- Orchestrated the development and publication of CyberArk Privileged Access Management (PAM) Architecture documentation in U.S. Bank's Architecture catalog, employing best practices in cybersecurity to fortify privileged account security, aligning with industry standards like SOC 2.
- Transformed IAM provisioning operations in Active Directory, CyberArk, and Unix/Linux servers through semi-automation using Python and PowerShell scripts, achieving a 90% reduction in manual workload, and ensuring LDAP and SAML protocol compliance, significantly elevating process efficiency and security posture.

Joint Fire Support Specialist
U.S. Army | Anoka, MN 2013 –2019

Education

Master of Science in Information Technology:
Enterprise Networks & Cloud Computing — Capella University, 2023

Bachelor of Business Administration (BBA) University of Northwestern - St. Paul, 2019

Certifications

Software Security Practitioner (SSP)	2021
Microsoft Certified: Azure Fundamentals	2021
Microsoft Certified: Azure Data Fundamentals	2022